Unified Tenant Isolation and QoS Detection in Kubernetes

Course Project for IT418 Cloud Computing

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PROBLEM STATEMENT

- Kubernetes is the backbone of modern cloud platforms and in almost every deployment, multiple tenants share the same cluster
- These tenants are separated by **RBAC policies**, **network rules**, and **resource quotas**. but these boundaries fail in two very different ways:

1. *Isolation violation*
2. *Fairness violation*

- Existing tools (Falco, Tetragon, kube-bench) catch security misconfigs but completely ignore the fairness problem. No tool unifies both.

OUR APPROACH

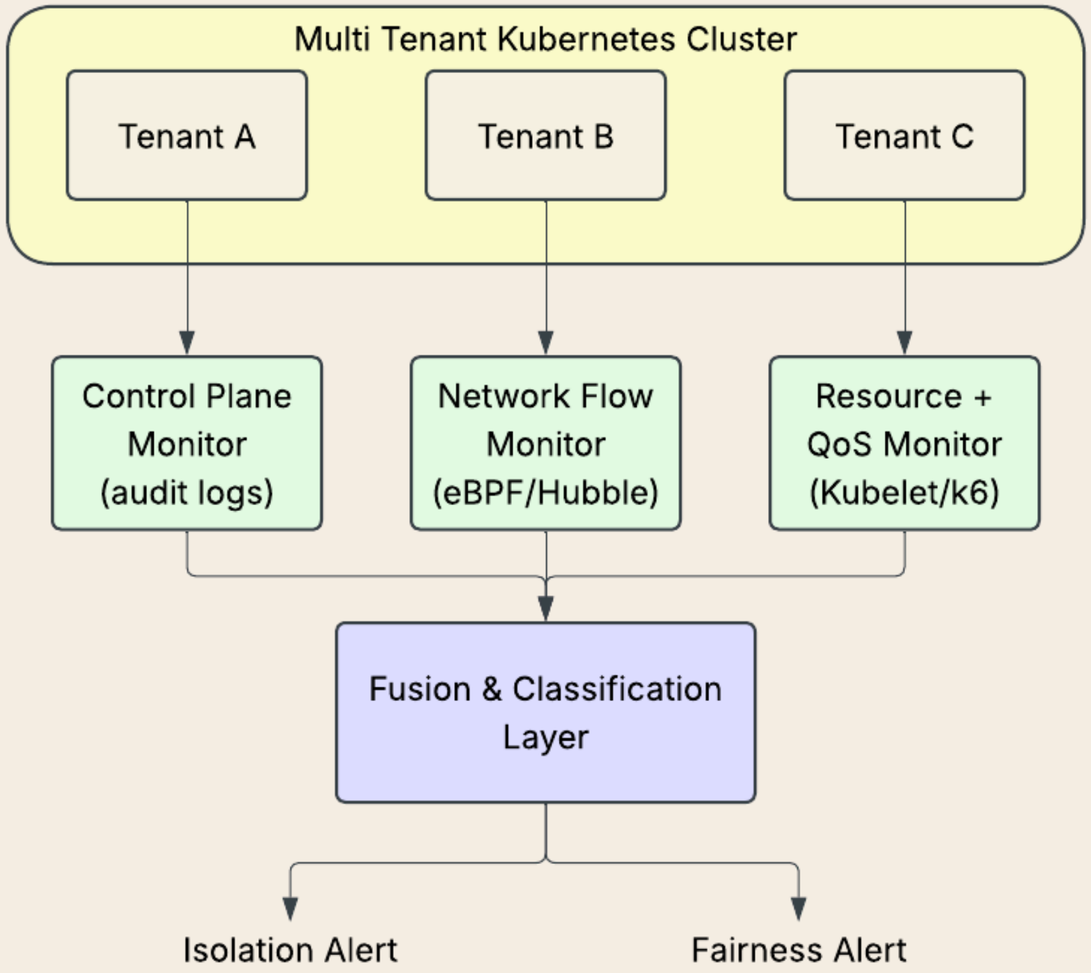
- Three-plane detection pipeline:

  Control plane: Kubernetes audit logs (who accessed what)

  Network plane: eBPF/Cilium/Hubble (traffic crossing tenant boundaries)

  Resource plane: kubelet/cAdvisor + k6 (CPU pressure + victim-side latency)
- All three feed into a fusion layer that correlates signals by namespace pair and time window, classifying each incident as an **isolation violation**, **fairness violation**, or both

SYSTEM ARCHITECTURE



All monitors write to a shared JSONL event stream; the fusion engine correlates by namespace pair within a 10-second window.

TESTBED SETUP

- Local kind cluster with Cilium/Hubble (eBPF CNI) and Kubernetes audit logging
- Demo FastAPI app deployed across 3 tenant namespaces, each isolated with:
  - RBAC: read-only ServiceAccount scoped to its own namespace
  - Network Policy: default-deny ingress + egress
  - Resource Quota: CPU, memory, and pod limits enforced
- Boundaries verified: cross-namespace API calls denied, cross-namespace traffic dropped, over-quota pods rejected

STATIC BASELINE

- Ran kube-bench (CIS Kubernetes 1.9) against the cluster

Category	PASS	FAIL	WARN
Master	40	7	12
etcd	7	0	0
Control Plane	1	0	4
Node	16	2	6
Policies	2	4	29
Total	66	13	51

- **29 WARN** in Policies, most tenant-boundary checks are Manual, kube-bench can't verify them at runtime
- Supplementary quota check: **3/3 PASS**

WORK DONE

- ***Control-plane monitor***: tails audit.log, flags cross-namespace API access
- ***Network-flow monitor***: eBPF via Hubble, detects traffic crossing tenant boundaries

AT

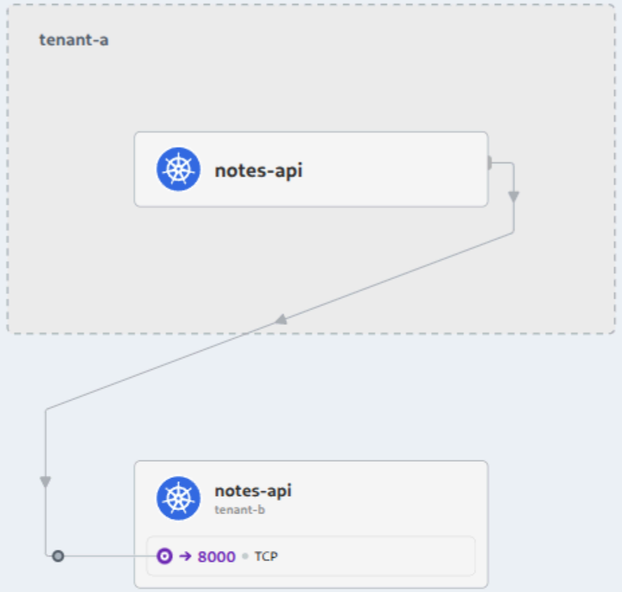
- ***Resource monitor***: kubelet/cAdvisor, flags CPU throttling + memory pressure
- ***QoS monitor***: k6 rolling p95 latency, emits breach when p95 > 1.5× baseline
- ***Fusion engine***: correlates events in 10s window, classifies alerts

tenant-a

Filter by: label key=val, ip=1.1.1.1, dns=google.com, identity=42, pod=frontend

Any verdict | Visual

16.9 flows/s • 3/3 nodes



Columns

Source Identity	Destination Identity	Destination Port	L7 info	Verdict	Timestamp
notes-api *tenant-a*	notes-api *tenant-b*	8000	—	forwarded	2026/08/30 19:48:25 (+…
notes-api *tenant-a*	notes-api *tenant-b*	8000	—	forwarded	2026/08/30 19:48:25 (+…
notes-api *tenant-a*	notes-api *tenant-b*	8000	—	forwarded	2026/08/30 19:48:25 (+…
notes-api *tenant-a*	notes-api *tenant-b*	8000	—	forwarded	2026/08/30 19:48:25 (+…
notes-api *tenant-a*	notes-api *tenant-b*	8000	—	forwarded	2026/08/30 19:48:25 (+…
notes-api *tenant-a*	notes-api *tenant-b*	8000	—	forwarded	2026/08/30 19:48:25 (+…
notes-api *tenant-a*	notes-api *tenant-b*	8000	—	forwarded	2026/08/30 19:48:22 (+0…
notes-api *tenant-a*	notes-api *tenant-b*	8000	—	forwarded	2026/08/30 19:48:22 (+0…
notes-api *tenant-a*	notes-api *tenant-b*	8000	—	forwarded	2026/08/30 19:48:22 (+0…
notes-api *tenant-a*	notes-api *tenant-b*	8000	—	forwarded	2026/08/30 19:48:22 (+0…

notes-api (tenant-a) 0.1.0 OAS 3.1

/openapi.json

default

GET /healthz Healthz

GET /whoami Whoami

GET /notes List Notes

POST /notes Create Note

GET /notes/{note_id} Get Note

DELETE /notes/{note_id} Delete Note

Schemas

HTTPValidationError › Expand all object

Note › Expand all object

NoteIn › Expand all object

ValidationError › Expand all object

FUTURE WORK

- Build and validate **8 injection scenarios** (5 isolation + 3 fairness) with automated inject/revert and ground-truth logging
- Run full evaluation sweep, compute precision, recall, false-positive rate, and detection latency AT
- **Ablation study**, fused pipeline vs. individual detectors to quantify value of multi-source fusion
- Live streaming mode for fusion engine, currently batch-oriented, extend to real-time alert emission
- Deploy on a real multi-node cluster to validate beyond single-host kind environment

THANK YOU